

STAT 516: Homework 6

Instructions: Please submit your homework as a single PDF on Gradescope. After uploading, please make sure each page is assigned to the correct question before submitting. Show your work clearly.

1. **(15 points)** Let (X, Y) have joint PDF

$$f_{X,Y}(x, y) = \begin{cases} 4xy, & 0 < x < 1, 0 < y < 1, \\ 0, & \text{otherwise.} \end{cases}$$

- (a) **(8 points)** Find the marginal PDFs $f_X(x)$ and $f_Y(y)$.
- (b) **(7 points)** Determine whether X and Y are independent.

2. **(13 points)** Let (X, Y) have joint PDF

$$f_{X,Y}(x, y) = \begin{cases} \frac{6}{5}(x^2 + y), & 0 < x < 1, 0 < y < 1, \\ 0, & \text{otherwise.} \end{cases}$$

Calculate

$$E[2X - Y + XY].$$

3. **(12 points)** Let X represent the age of an insured automobile involved in an accident. Let Y represent the length of time the owner has insured the automobile at the time of the accident. Suppose X and Y have joint PDF

$$f_{X,Y}(x, y) = \begin{cases} \frac{1}{64}(10 - xy^2), & 2 \leq x \leq 10, 0 \leq y \leq 1, \\ 0, & \text{otherwise.} \end{cases}$$

Calculate the expected age of an insured automobile involved in an accident, $E[X]$.

4. **(12 points)** Let X and Y be continuous random variables with joint PDF

$$f_{X,Y}(x, y) = \begin{cases} \frac{1}{4}, & 0 \leq x \leq 2, x - 2 \leq y \leq x, \\ 0, & \text{otherwise.} \end{cases}$$

Calculate

$$E[X^3Y].$$

5. **(12 points)** The profit of a new product is given by

$$Z = 3X - Y - 5.$$

Assume X and Y are independent random variables with

$$\text{Var}(X) = 1 \quad \text{and} \quad \text{Var}(Y) = 2.$$

Calculate $\text{Var}(Z)$.

6. **(12 points)** The profit of a new product is given by

$$Z = 5X - 2Y - 3.$$

Assume

$$\text{Var}(X) = 1, \quad \text{Var}(Y) = 2, \quad \text{Cov}(X, Y) = -1.$$

Calculate $\text{Var}(Z)$.

7. **(12 points)** An actuary analyzes a company's annual personal auto claim, M , and annual commercial auto claim, N . The analysis reveals that

$$\text{Var}(M) = 1600, \quad \text{Var}(N) = 900,$$

and the correlation between M and N is

$$\rho(M, N) = 0.64.$$

Calculate

$$\text{Var}(M + N).$$

8. **(12 points)** Let X and Y be continuous random variables with joint PDF

$$f_{X,Y}(x, y) = \begin{cases} \frac{8}{3}xy, & 0 \leq x \leq 1, \ x \leq y \leq 2x, \\ 0, & \text{otherwise.} \end{cases}$$

Calculate

$$\text{Cov}(X, Y).$$